

Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#4)

Use of Venus Simulator for Register Level Execution of RISC-V Arithmetic and Memory Instructions

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Date performed: 8th September, 2026

MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____

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Version: 1.0

Verified by: Dr. Nasir Ali Shah

Updated: Fall 2026

Task 1:

Description:

We had to write the RISC-V assembly code for the given C programs using A = 5 and B = 4. We then identified the different fields of each instruction according to its instruction format and represented the values of these fields in decimal.

Code:

```
.data
A: .word 5
B: .word 4

.text
lw t0,A
lw t1,B
add s2,t0,t1
sub s3,t0,t1
mul s4,t0,t1
mul s5,t0,t0
```

Output:

PC	Machine Code	Basic Code	Original Code	sp (x2)	2147483616
0x0	0x10000297	auipc x5 65536	lw t0,A	gp (x3)	268435456
0x4	0x0002A283	lw x5 0(x5)	lw t0,A	tp (x4)	0
0x8	0x10000317	auipc x6 65536	lw t1,B	t0 (x5)	268435456
				t1 (x6)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	lw t0,A
0x4	0x0002A283	lw x5 0(x5)	lw t0,A
0x8	0x10000317	auipc x6 65536	lw t1,B
0xc	0xFFC32303	lw x6 -4(x6)	lw t1,B

0x4	0x0002A283	lw x5 0(x5)	lw t0,A
0x8	0x10000317	auipc x6 65536	lw t1,B
0xc	0xFFC32303	lw x6 -4(x6)	lw t1,B
0x10	0x00628933	add x18 x5 x6	add s2,t0,t1

0x8	0x10000317	auipc x6 65536	lw t1,B
0xc	0xFFC32303	lw x6 -4(x6)	lw t1,B
0x10	0x00628933	add x18 x5 x6	add s2,t0,t1
0x14	0x406289B3	sub x19 x5 x6	sub s3,t0,t1
0x18	0x02628A33	mul x20 x5 x6	mul s4,t0,t1
0x1c	0x02528AB3	mul x21 x5 x5	mul s5,t0,t0

0xc	0xFFC32303	lw x6 -4(x6)	lw t1,B
0x10	0x00628933	add x18 x5 x6	add s2,t0,t1
0x14	0x406289B3	sub x19 x5 x6	sub s3,t0,t1
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0x18	0x02628A33	mul x20 x5 x6	mul s4,t0,t1
0x1c	0x02528AB3	mul x21 x5 x5	mul s5,t0,t0

Task 2:

Description:																				
Converted the following C code into RISC-V assembly language. Using appropriate registers for the variables. m = (a - c) + (d + e) * f Where t0=a t1=c t2=d t3=e t4=f t5=m																				
Code:																				
.text #t0=a t1=c t2=d t3=e t4=f t5=m sub t5,t0,t1 add t6,t2,t3 mul t6,t6,t4 add t5,t5,t6																				
Output:																				
<table><tr><th>PC</th><th>Machine Code</th><th>Basic Code</th><th>Original Code</th></tr><tr><td>0x0</td><td>0x40628F33</td><td>sub x30 x5 x6</td><td>sub t5,t0,t1</td></tr><tr><td>0x4</td><td>0x01C38FB3</td><td>add x31 x7 x28</td><td>add t6,t2,t3</td></tr><tr><td>0x8</td><td>0x03DF8FB3</td><td>mul x31 x31 x29</td><td>mul t6,t6,t4</td></tr><tr><td>0xc</td><td>0x01FF0F33</td><td>add x30 x30 x31</td><td>add t5,t5,t6</td></tr></table>	PC	Machine Code	Basic Code	Original Code	0x0	0x40628F33	sub x30 x5 x6	sub t5,t0,t1	0x4	0x01C38FB3	add x31 x7 x28	add t6,t2,t3	0x8	0x03DF8FB3	mul x31 x31 x29	mul t6,t6,t4	0xc	0x01FF0F33	add x30 x30 x31	add t5,t5,t6
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Task 3:

Description:
Here we wrote RISC-V assembly language program to perform AND, OR, XOR on the following numbers using Venus. 0x f f f 0x f 0 f Using and or xor instructions

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Code:

```
li t0, 0xfff
li t1, 0xf0f
and t2,t0,t1
or t3,t0,t1
xor t4,t0,t1
```

Output:

tp (x4) 0x00000000

t0 (x5) 0x00000FFF

t1 (x6) 0x00000000

tp (x4) 0x00000000

t0 (x5) 0x00000FFF

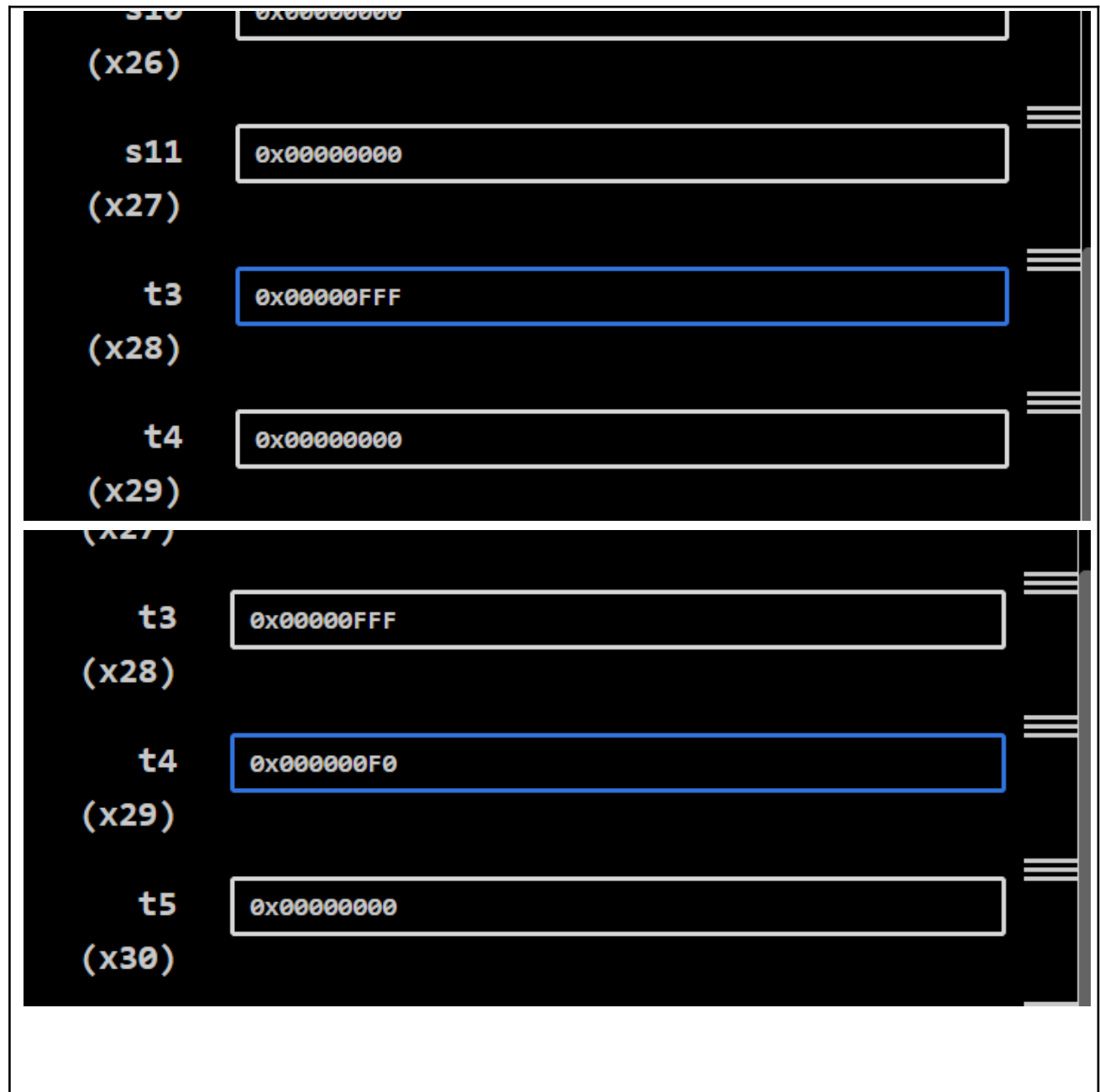
t1 (x6) 0x00000F0F

t0 (x5) 0x00000FFF

t1 (x6) 0x00000F0F

t2 (x7) 0x00000F0F

s0 (x8) 0x00000000



Task 4:

Description:

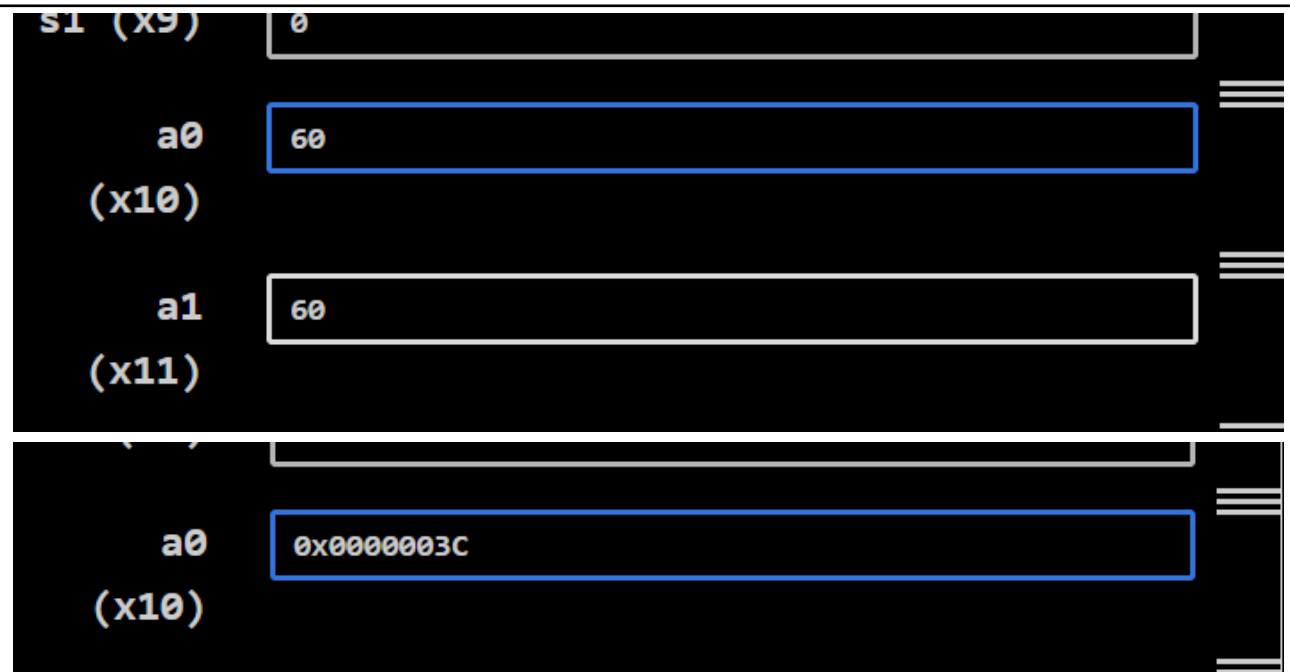
We had to test the given code and check the register values to verify that it was working as intended. To run the code successfully, a main program was added to initialize the variables with values, and the labels were modified to make the code compatible with the Venus Simulator. The final value was also displayed in the console output at the end of the program.

Code:

```
.data
array: .word 10, 20, 30

.text
main:
    la a0, array
    li a1, 3
    jal ra, arraysum
j done
    # When we get here, a0 should be 60
arraysum:
    # a0 = int a[]
    # a1 = int size
    # t0 = ret
    # t1 = i
    li t0, 0 # ret = 0
    li t1, 0 # i = 0
loop: # For loop
    bge t1, a1, finish # if i >= size, break
    slli t2, t1, 2 # Multiply i by 4 (1 << 2 = 4)
    add t2, a0, t2 # Update memory address
    lw t2, 0(t2) # Dereference address to get integer
    add t0, t0, t2 # Add integer value to ret
    addi t1, t1, 1 # Increment the iterator
    j loop # Jump back to start of loop (1 backwards)
finish:
    mv a0, t0 # Move t0 (ret) into a0
    ret # Return via return address register
done:
    mv a1, a0
    li a0, 1
    ecall
```

Output:



Task 5:

Description:

In this we Wrote RISC-V assembly code to write a table of two. then initialized table of two in the data section using .data in Venus. After this, loaded table values of two into different registers using lw command. Since the temporary registers only go up to t6, the rest of the values were stored in s1 onwards,

Code:

```
.data
tableoftwo: .word 2,4,6,8,10,12,14,16,18,20

.text
main:
la t0, tableoftwo
lw t1, 0(t0)
lw t2, 4(t0)
lw t3, 8(t0)
lw t4, 12(t0)
lw t5, 16(t0)
lw t6, 20(t0)
lw s1, 24(t0)
```

lw s2, 28(t0)

lw s3, 32(t0)

lw s4, 36(t0)

Output:

t1 (x6)

0x00000002

t2

0x00000004

(x7)

t3

0x00000006

(x28)

t4

0x00000008

(x29)

t5

0x0000000A

(x30)

